

Project Planning

This document covers the work breakdown structure, sprint plan, team roles, risk register, and resource plan for the OptiCrop project.

Sprint Timeline Gantt Chart

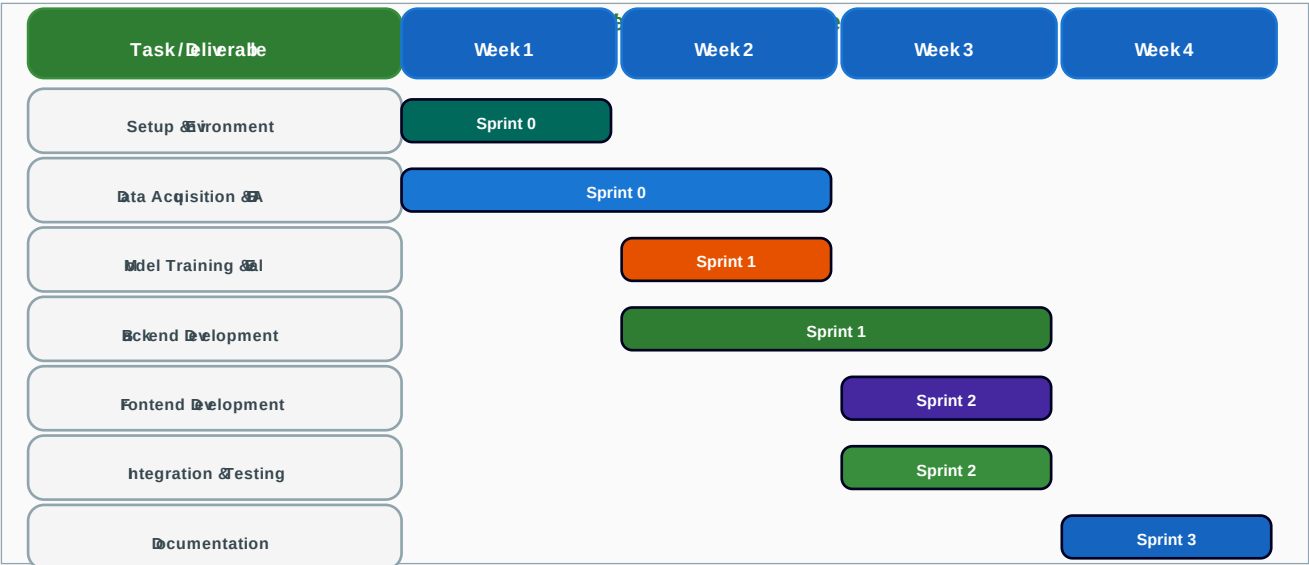


Figure 5: 4-Week Sprint Gantt Chart

Work Breakdown Structure (WBS)

WBS ID	Deliverable	Subtasks	Owner	Duration
1	Project Initiation	Kick-off, team setup, tool installation	All	3 days
2	Data Acquisition	Download Kaggle dataset, verify schema, sample review	Data Lead	2 days
3	Exploratory Analysis	Distribution plots, correlation heatmap, outlier checks	Data Lead	3 days
4	Model Development	Train 5 models, tune hyperparams, select best	ML Lead	5 days
5	Model Persistence	Pickle model + scaler, write unit tests	ML Lead	2 days
6	Backend Development	Flask app, routes, predictor.py, preprocess.py	Backend Dev	5 days
7	Frontend Development	HTML templates, Bootstrap styling, JS validation	Frontend Dev	4 days

8	Integration & Testing	End-to-end test, fix bugs, performance test	All	3 days
9	Documentation	README, user guide, code comments, this document set	All	4 days
10	Demonstration	Slides, demo video, live session preparation	All	3 days

RiskRegister

Risk	Probability	Impact	Mitigation
Model accuracy below 90%	Low	High	Try ensemble methods; increase training data via augmentation
Dataset quality issues (null values)	Medium	Medium	Pandas fillna + IQR outlier removal during preprocessing
Flask deployment environment issues	Low	Medium	Provide detailed venv setup in README; Docker fallback
Team member unavailability	Medium	Medium	Each section cross-trained; documented sprint responsibilities
Scope creep (feature additions)	High	Low	Strict MoSCoW; defer nice-to-haves to Phase 2 backlog

Team Roles

Member	Role	Primary Responsibilities
Member 1	Team Lead & Backend Developer	Flask architecture, sprint planning, GitHub management, risk tracking
Member 2	ML Engineer	Dataset EDA, model training, model.pkl, scaler.pkl, metrics
Member 3	Frontend Developer	HTML templates, Bootstrap UI, JS validation, mobile responsiveness
Member 4	QA Engineer & Tech Writer	pytest suite, performance tests, all 18 documents, README